MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GATE, AND, DIGITAL, KEY CODE DQK

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic AND gate module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.051 failures per million operating hours for class I or 0.085 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be DQK.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/243-1 for class I and M28787/243-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

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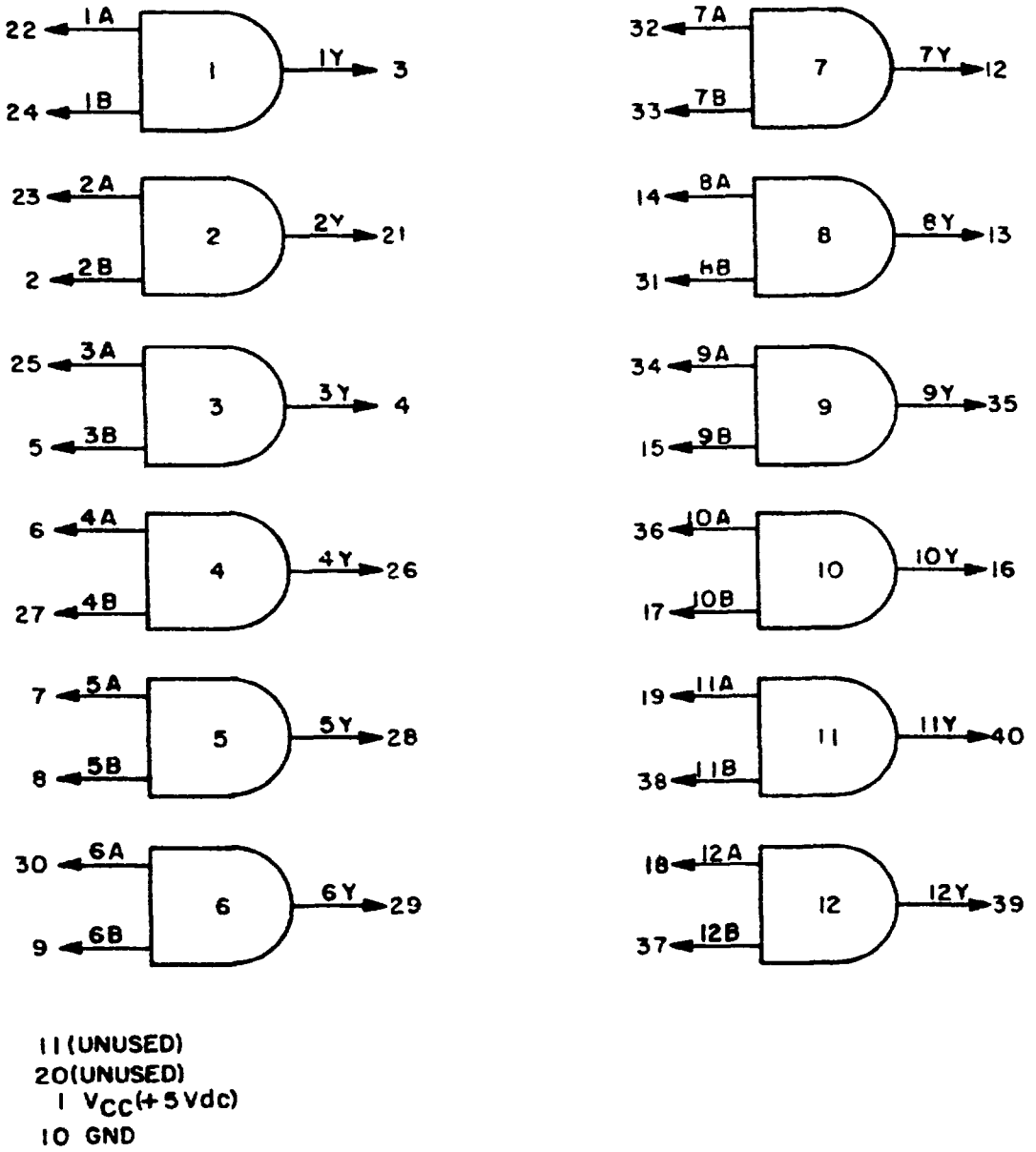


FIGURE 1. DQK module function diagram, AND gate.

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* TABLE I. Module electrical requirements

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1	Isolation Initial End-of-life	III III	S S	10 1	∞ ∞	10 1		10 1	∞ ∞	10 1	∞ ∞	10 1	∞ ∞	MΩ MΩ
A2-A4	Isolation Initial End-of-life	III III	X X	10 1	∞ ∞	10 1	∞ ∞	10 1	∞ ∞	10 1	∞ ∞	10 1	∞ ∞	MΩ MΩ
B1-B12	V_{OUT} (low loaded)	IV	S	10	800	10	800	10	800	10	800	10	800	mV
C1-C24	Input breakdown current	V	S	0	1	0	1	0	1	0	1	0	1	mA
D1-D24	Input clamp diode voltage	VI	S	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
E1-E12	Output short circuit current	VII	X	-20	-55	-20	-55	-20	-55	-20	-55	-20	-55	mA
F1-F12	Output voltage, high level	VIII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
G1-G24	Output voltage, low level	IX	X	0	400	0	400	0	400	0	400	0	400	mV
H1-H24	Input current, high level	X	X	0	40	0	40	0	40	0	40	0	40	µA
J1-J24	Input current, low level	XI	X	-0.10	-1.73	-0.10	-1.73	-0.10	-1.73	-0.10	-1.73	-0.10	-1.73	mA
K1	Power supply current, low level	XII	S	1	99	1	99	1	99	1	99	1	99	mA
K2	Power supply current, high level	XII	S	1	63	1	63	1	63	1	63	1	63	mA
L1-L12	Transition time (t_{TLH}), C_L = 50 pF	XIII	S	3	45	3	50	3	50	3	50	3	50	ns
L13-L24	Transition time (t_{TLH}), C_L = 150 pF	XIII	Q	3	70	3	75	3	75	3	75	3	75	ns
M1-M12	Transition time (t_{THL}), C_L = 50 pF	XIV	S	4	20	4	25	4	25	4	25	4	25	ns
M13-M24	Transition time (t_{THL}), C_L = 150 pF	XIV	Q	4	30	4	35	4	35	4	35	4	35	ns
N1-N24	Propagation delay time (t_{PLH}), C_L = 50 pF	XV	S	3	31	3	36	3	36	3	36	3	36	ns

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
N25-N48	Propagation delay time (t_{PLH}), C_L = 150 pF	XV	Q	3	41	3	46	3	46	3	46	3	46	ns
P1-P24	Propagation delay time (t_{PHL}), C_L = 50 pF	XVI	S	3	22	3	27	3	27	3	27	3	27	ns
P25-P48	Propagation delay time (t_{PHL}), C_L = 150 pF	XVI	Q	3	32	3	37	3	37	3	37	3	37	ns
Q1-Q12	Output crosstalk, high level	XVII	Q	0	400	0	400	0	400	0	400	0	400	mV
R1-R12	Output crosstalk, low level	XVIII	Q	0	800	0	800	0	800	0	800	0	800	mV
S	Power supply transient amplitude	XIX	Q	0	400	0	400	0	400	0	400	0	400	mV
T	Power supply filter capactiance	XX	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF
U1-U24	Input capacitance	XXI	Q	2	15	2	15	2	15	2	15	2	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through U shall be performed in any order with the following exceptions If test conditions A, B, C, or E are part of a sequence of tests, the order of testing shall be as follows: A, B, C, E where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through U in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through T shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition T, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through P shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through U shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5 PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6 NOTES

6.1 Definitions Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

* 6.4 Module description. The following electrical parameters reflect the 0° to 60°C and -55° to 100°C end-of-life limits from table I.

Maximum power dissipation	545 mW (outputs not loaded)
Supply voltage	5.0 V ±10%

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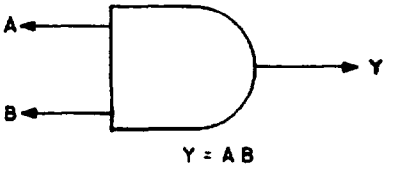
Input current		
Low level	(V_{IL} = 0.0 V)	-1 73 mA maximum
	(V_{IL} = 0.4 V)	-1.6 mA maximum
High level	(V_{IH} = 2.4 V)	40 µA maximum
	(V_{IH} = 5.5 V)	1.0 mA maximum
Input voltage		
Low level		-0.5 V minimum, 0.8 V maximum
High level		2.0 V minimum, V_{CC} maximum
Output voltage		
Low level	(V_{OL} = 16.0 mA)	0.4 maximum
High level	(I_{OH} = 0.8 mA)	2.4 V maximum
Output drive		
Low level	(V_{OL} = 0.4 V)	16.0 mA
High level	(V_{OH} = 2.4 V)	-0.8 mA

* 6.5 Module function. This module contains twelve independent 2-input, and totem-pole output AND gates for standard TTL positive logic.

6.5.1 Input to output relationship. Each AND gate has the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship.

Input		Output
A	B	Y
L	L	L
L	H	L
H	L	L
H	H	H



6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

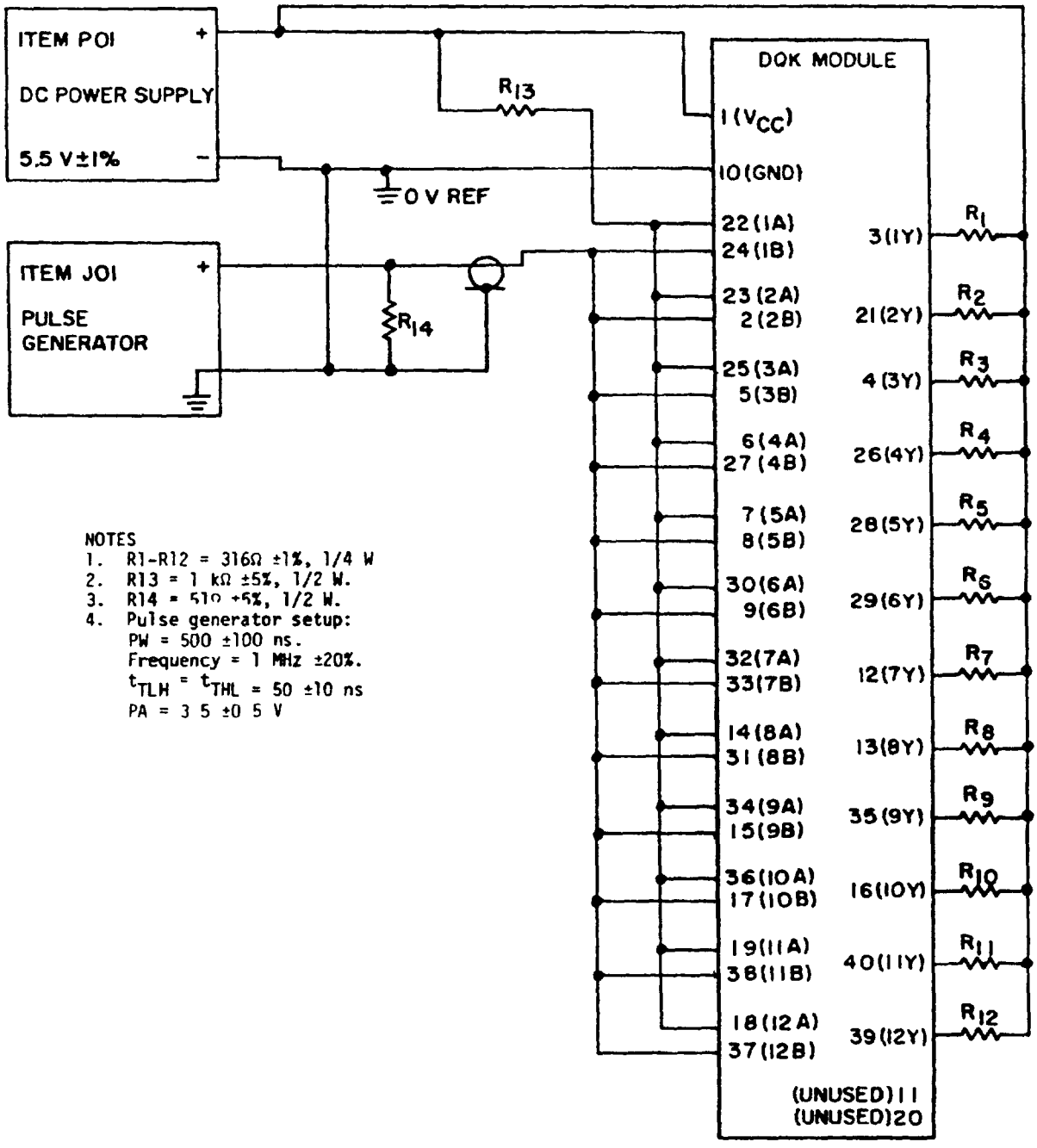
6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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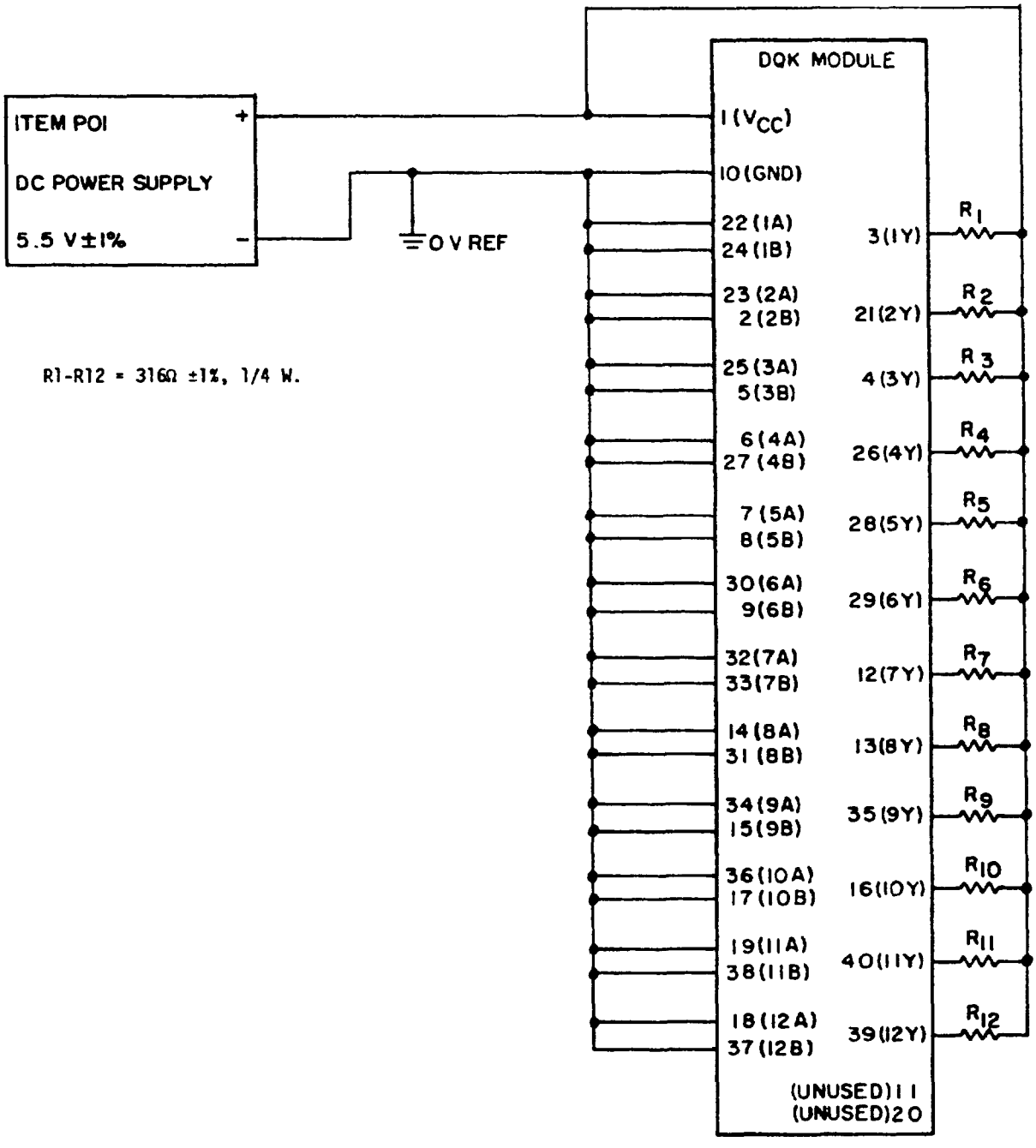
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NOTES
1. R1-R12 = 316Ω ±1%, 1/4 W
2. R13 = 1 kΩ ±5%, 1/2 W.
3. R14 = 510 ±5%, 1/2 W.
4. Pulse generator setup:
   PW = 500 ±100 ns.
   Frequency = 1 MHz ±20%.
   $t_{TLH} = t_{THL}$ = 50 ±10 ns
   PA = 3 5 ±0 5 V

FIGURE 2 Operating temperature and life test circuit.

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R1-R12 = 316Ω ±1%, 1/4 W.

FIGURE 3 Component temperature test

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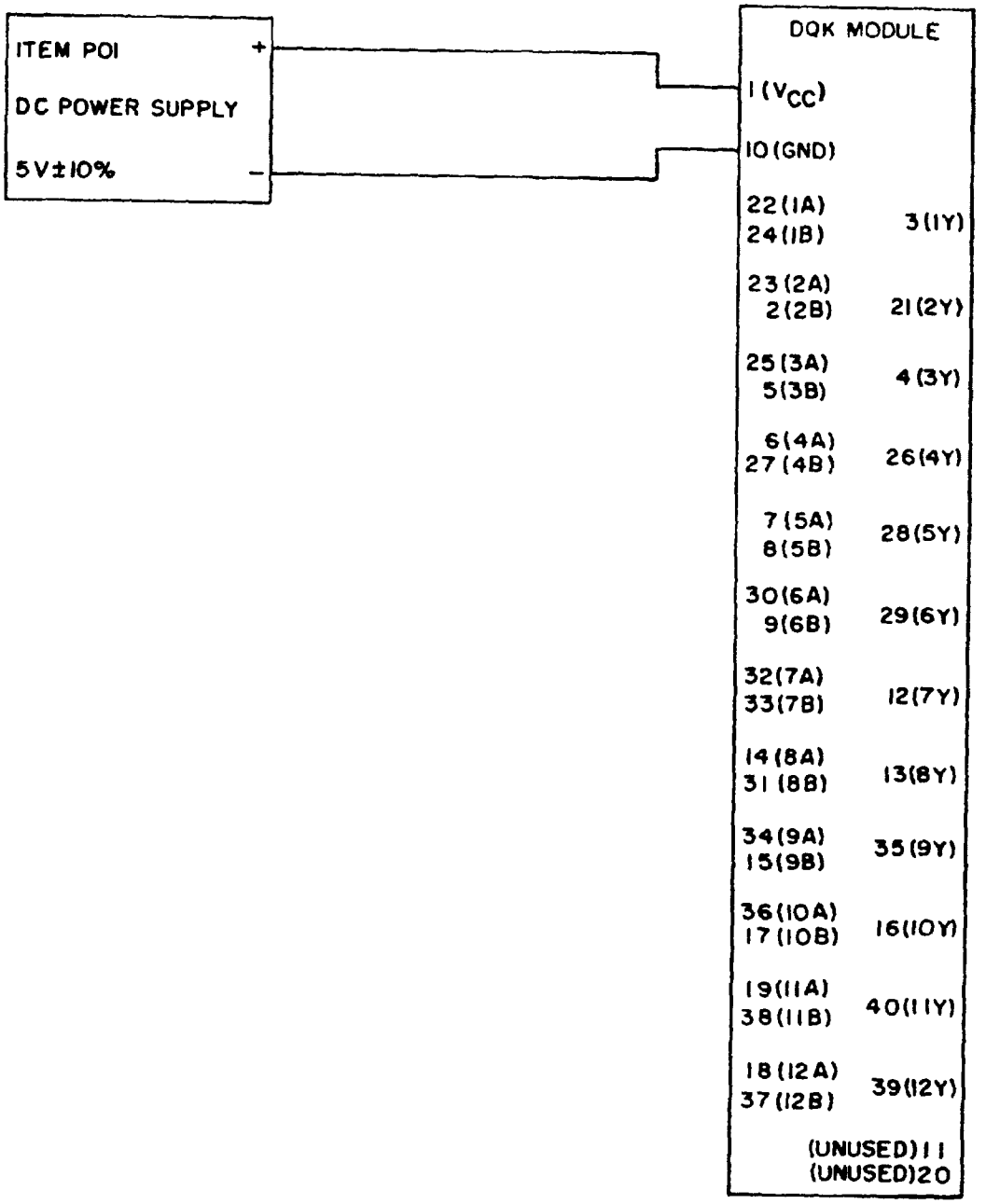


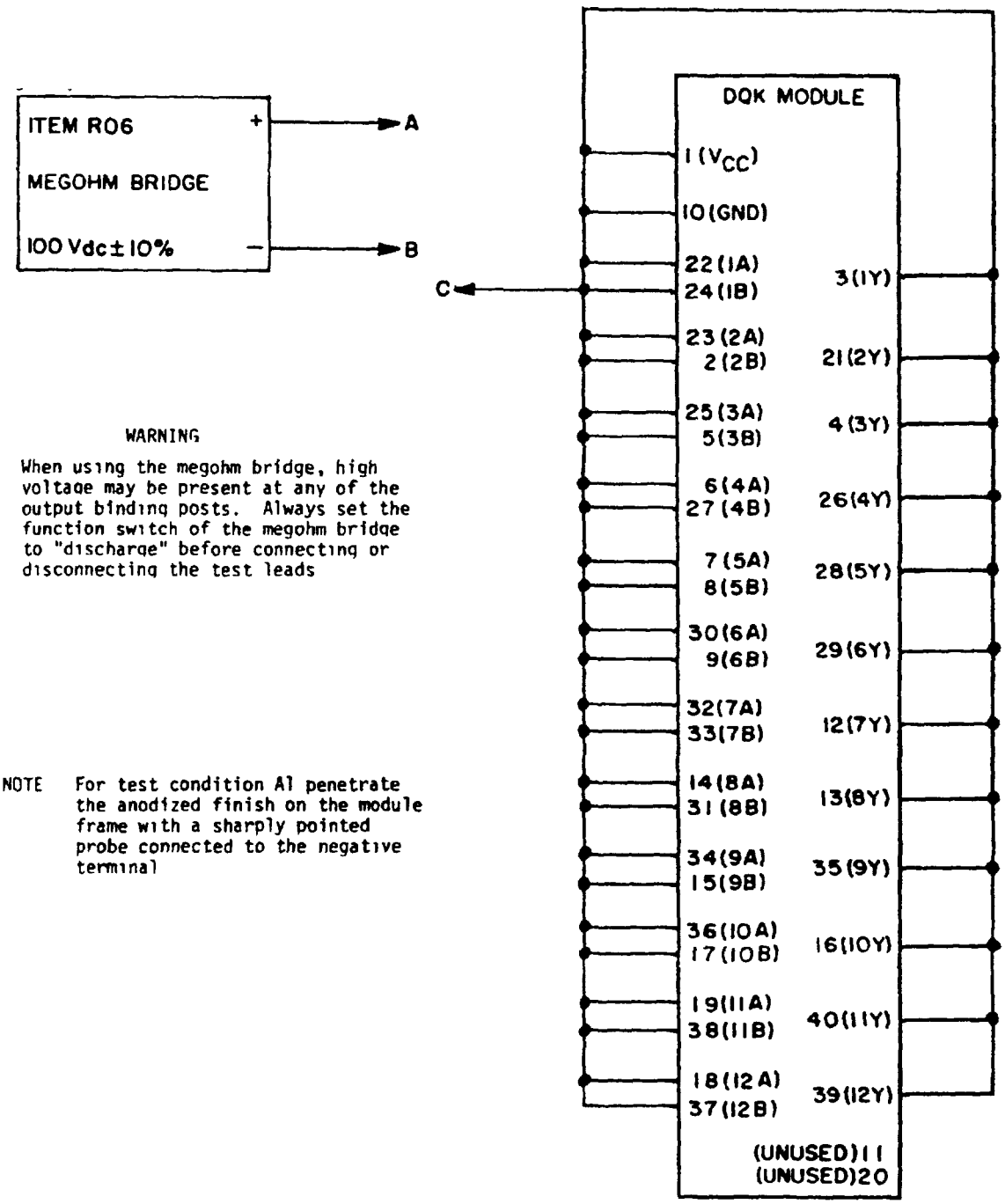
FIGURE 4 Vibration test circuit.

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* TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation	C, 11, 20	Frame	10	∞	MΩ	5
A2	↓	11	20	↓	↓	↓	↓
A3		C	11				
A4		C	20				

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WARNING

When using the megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch of the megohm bridge to "discharge" before connecting or disconnecting the test leads

NOTE For test condition A1 penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal

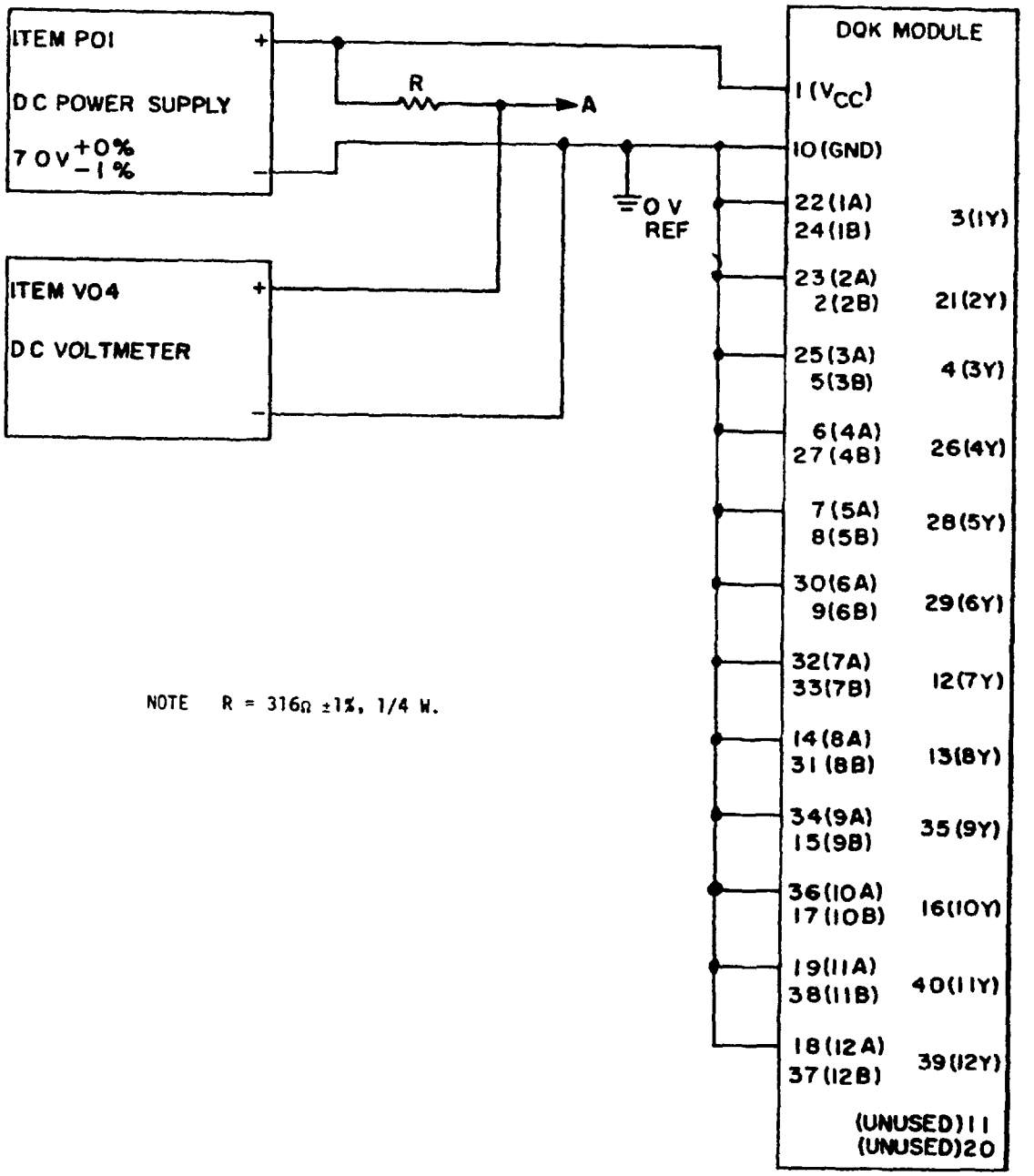
FIGURE 5 Isolation test circuit for table III

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* TABLE IV. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect A to pin	Initial limits 25 ±5°C ambient Min	Initial limits 25 ±5°C ambient Max	Unit	Test figure
B1	V_{OUT} (low loaded)	3	10	800	mV	6
B2		21				
B3		4				
B4		26				
B5		28				
B6		29				
B7		12				
B8		13				
B9		35				
B10		16				
B11		40				
B12		39				

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NOTE R = 316Ω ±1%, 1/4 W.

FIGURE 6. V_{OUT} (low-loaded) test circuit for table IV.

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TABLE V. Input breakdown current test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	Input breakdown current	22	24	0	1	mA	7
C2		24	22				
C3		23	2				
C4		2	23				
C5		25	5				
C6		5	25				
C7		6	27				
C8		27	6				
C9		7	8				
C10		8	7				
C11		30	9				
C12		9	30				
C13		32	33				
C14		33	32				
C15		14	31				
C16		31	14				
C17		34	15				
C18		15	34				
C19		36	17				

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TABLE V Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C20	Input breakdown current	17	36	0	1	mA	7
C21	↓	19	38	↓	↓	↓	↓
C22		38	19				
C23		18	37				
C24	↓	37	18	↓	↓	↓	↓

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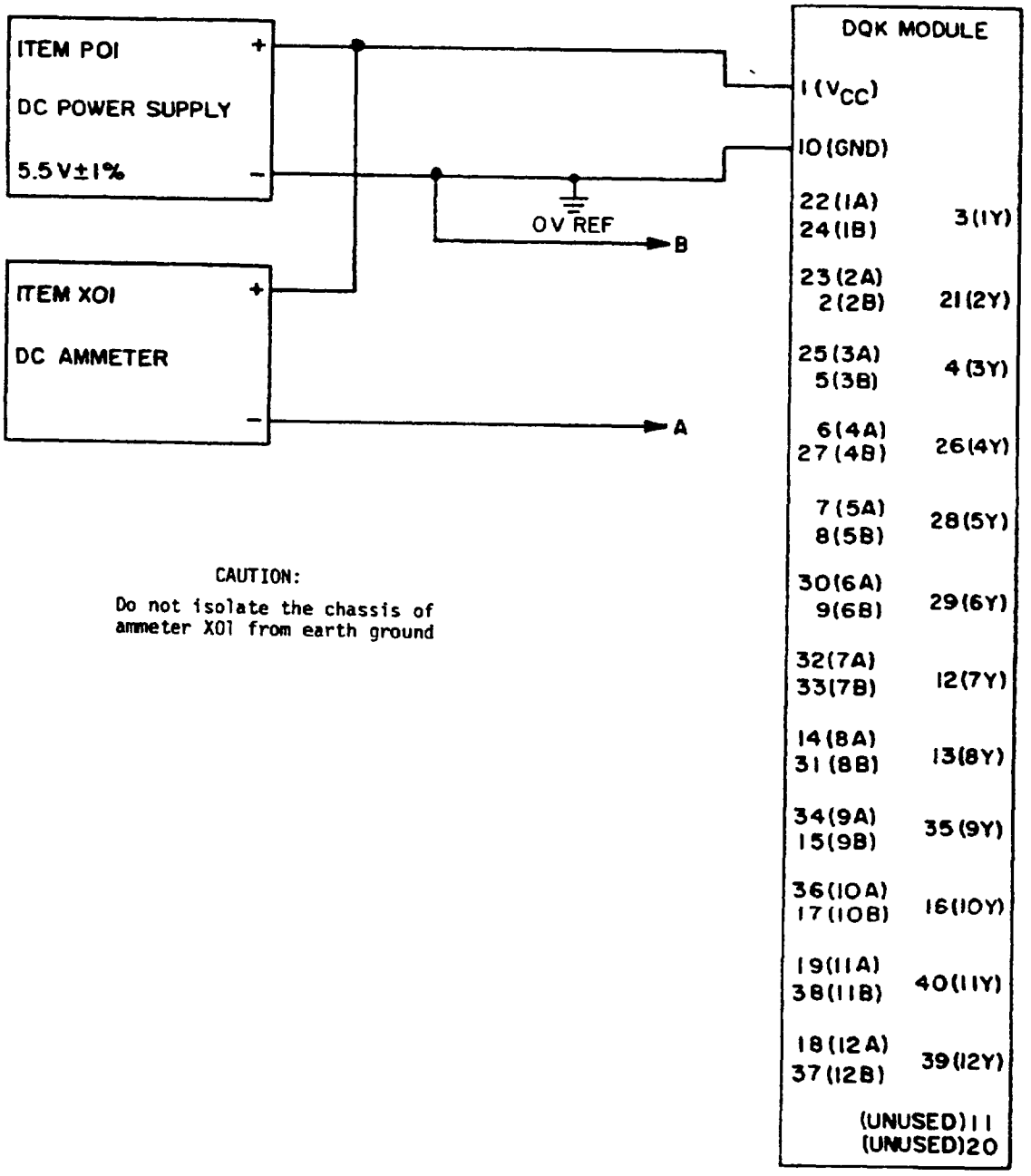


FIGURE 7 Input breakdown current test circuit for table V.

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TABLE VI. Input clamp diode voltage test.

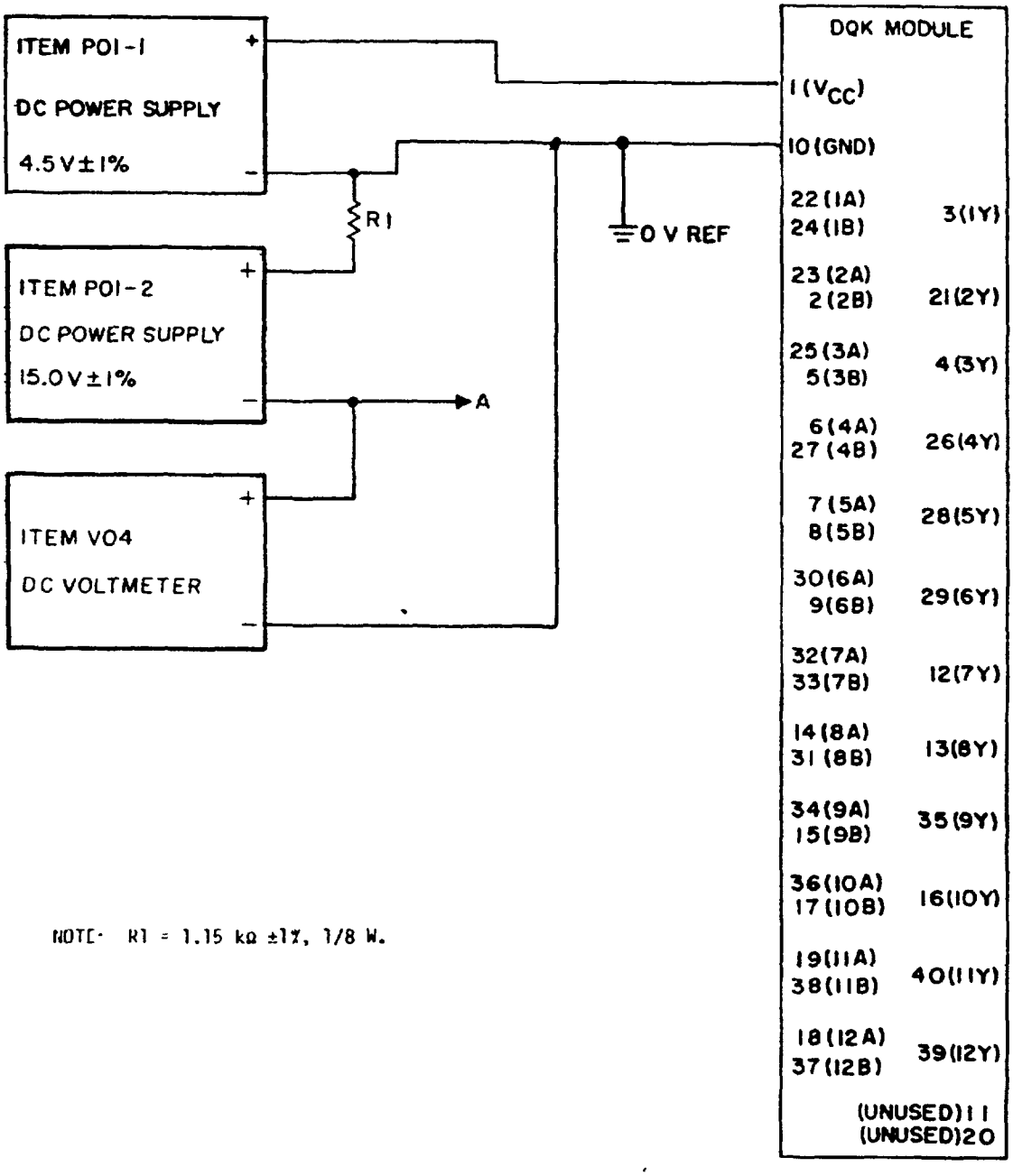
Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
D1	Input clamp diode voltage	22	-0.4	-1.5	V	8
D2		24				
D3		23				
D4		2				
D5		25				
D6		5				
D7		6				
D8		27				
D9		7				
D10		8				
D11		30				
D12		9				
D13		32				
D14		33				
D15		14				
D16		31				
D17		34				
D18		15				
D19		36				

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TABLE VI. Input clamp diode voltage test - Continued.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
D20	Input clamp diode voltage	17	-0.4	-1.5	V	8
D21	↓	19	↓	↓	↓	↓
D22		38				
D23		18				
D24		37				

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NOTE: R1 = 1.15 kΩ ±1%, 1/8 W.

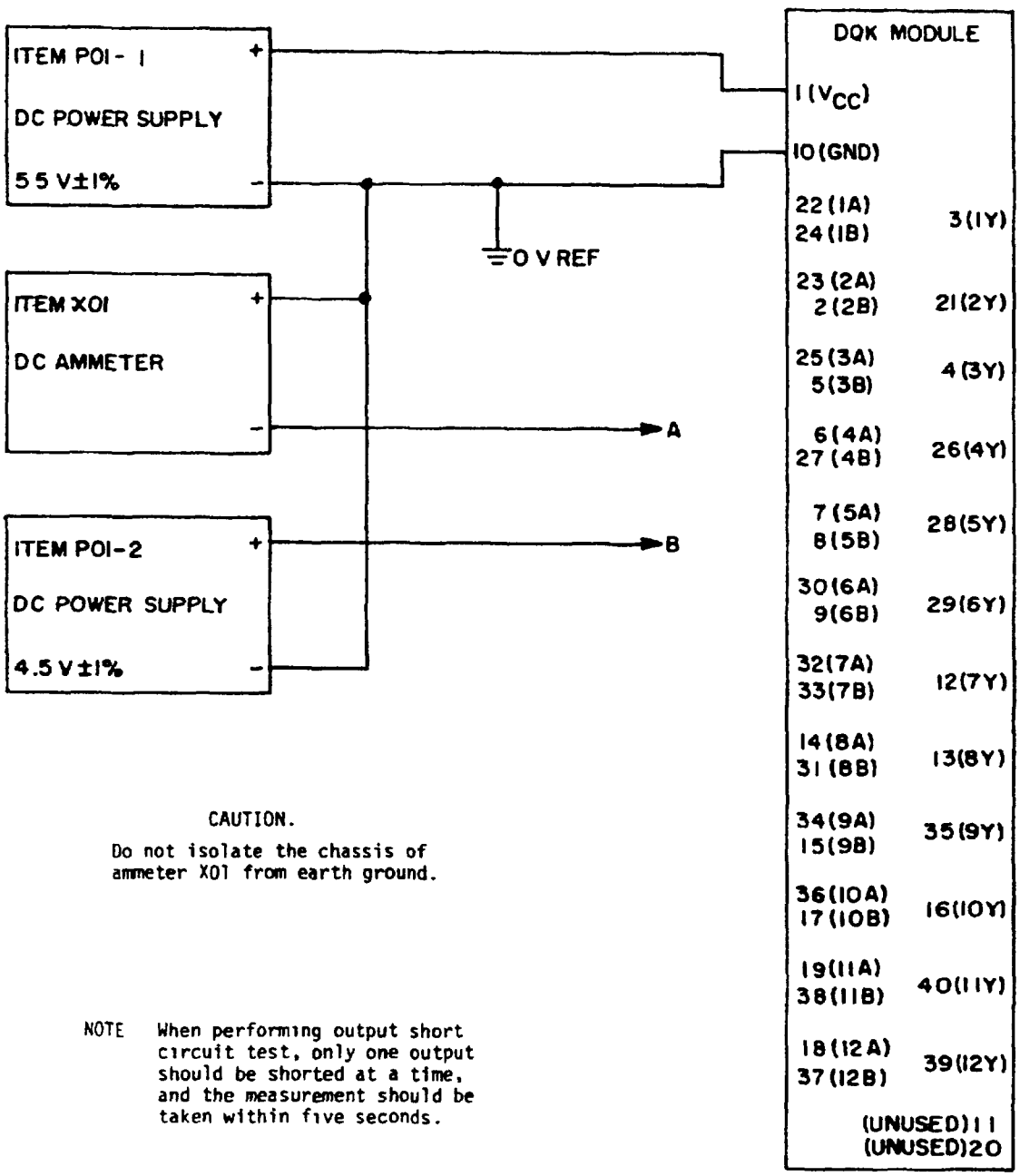
* FIGURE 8. Input clamp diode voltage test circuit for table VI.

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TABLE VII. Output short circuit current test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E1	Output short circuit current	3	22, 24	-20	-55	mA	9
E2		21	23, 2				
E3		4	25, 5				
E4		26	6, 27				
E5		28	7, 8				
E6		29	30, 9				
E7		12	32, 33				
E8		13	14, 31				
E9		35	34, 15				
E10		16	36, 17				
E11		40	19, 38				
E12		39	18, 37				

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CAUTION.

Do not isolate the chassis of ammeter X01 from earth ground.

NOTE When performing output short circuit test, only one output should be shorted at a time, and the measurement should be taken within five seconds.

FIGURE 9. Output short circuit test circuit for table VII.

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* TABLE VIII. Output voltage (high-level) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F1	Output voltage, high level	Open	22, 24	3	2.4	4.5	V	10
F2	↓	↓	23, 2	21	↓	↓	↓	↓
F3			25, 5	4				
F4			6, 27	26				
F5			7, 8	28				
F6			30, 9	29				
F7			32, 33	12				
F8			14, 31	13				
F9			34, 15	35				
F10			36, 17	16				
F11			19, 38	40				
F12			18, 37	39				

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* TABLE IX. Output voltage (low-level) test.

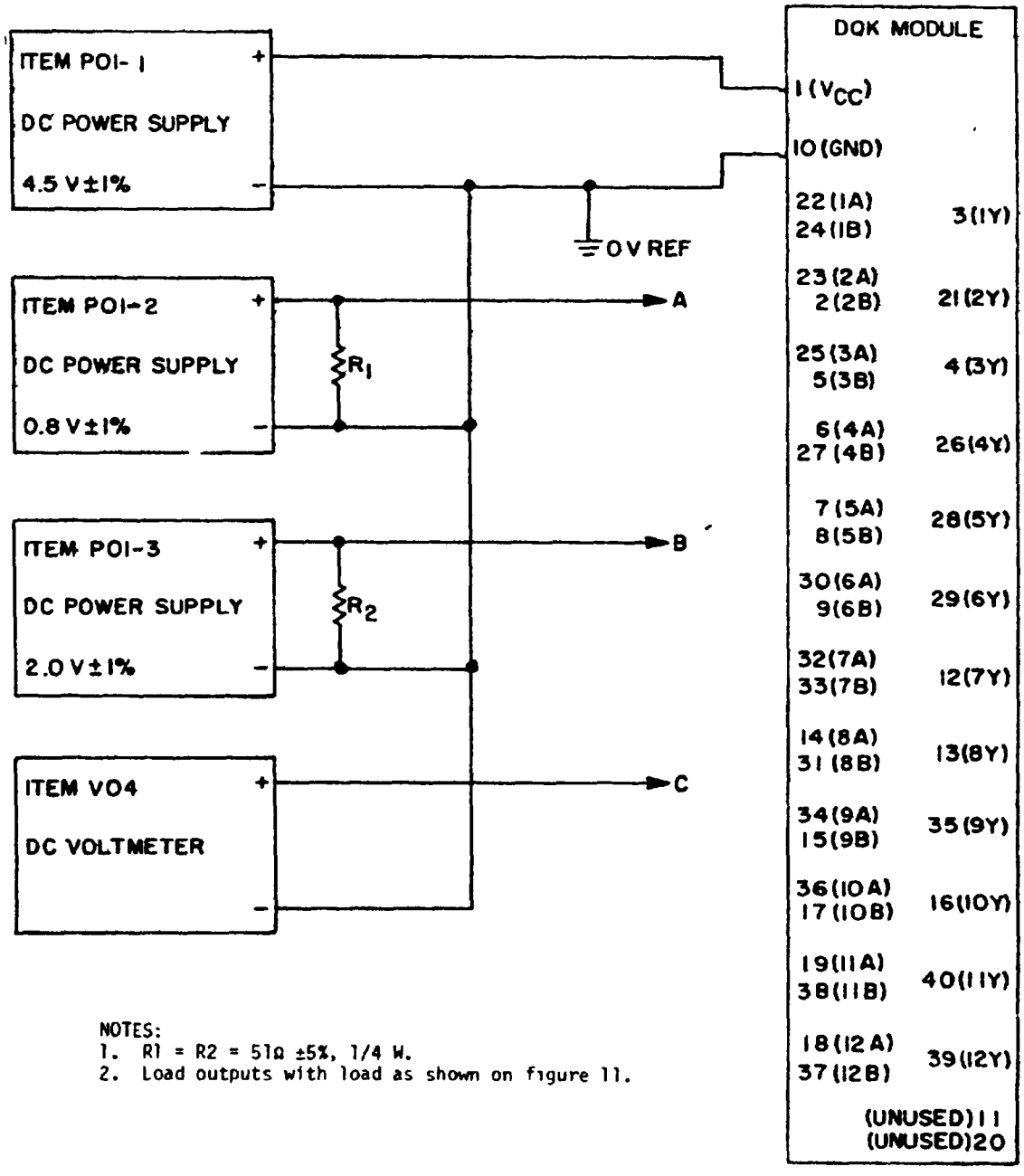
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G1	Output voltage, low level	22	24	3	0	400	mV	10
G2	↓	23	2	21	↓	↓	↓	↓
G3		25	5	4				
G4		6	27	26				
G5		7	8	28				
G6		30	9	29				
G7		32	33	12				
G8		14	31	13				
G9		34	15	35				
G10		36	17	16				
G11		19	38	40				
G12		18	37	39				
G13		24	22	3				
G14		2	23	21				
G15		5	25	4				
G16		27	6	26				
G17		8	7	28				
G18	↓	9	30	29	↓	↓	↓	↓

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* TABLE IX. Output voltage (low-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G19	Output voltage, low level	33	32	12	0	400	mV	10
G20	↓	31	14	13	↓	↓	↓	↓
G21		15	34	35				
G22		17	36	16				
G23		38	19	40				
G24		37	18	39				

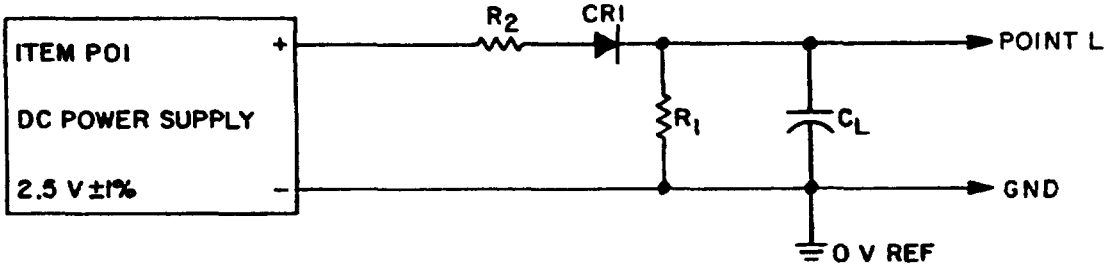
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/4 W.
2. Load outputs with load as shown on figure 11.

* FIGURE 10 Output voltage (high-level and low-level) test circuit for tables VIII and IX.

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NOTES:
1. CR1 = 1N4150 or equivalent.
2. R1 = 3.01 kΩ ±1%, 1/8 W.
3. R2 = Select value for current flow of 16 mA ±5% out of point L with point L held at 0.4 V.
4. C_L = 50 ±15 pF (including probe and parasitic capacitance) for test conditions L1-L12, M1-M12, N1-N24, P1-P24, Q1-Q12 and R1-R12.
5. C_L = 150 ±15 pF (including probe and parasitic capacitance) for test conditions L13-L24, M13-M24, N25-N48 and P25-P48.
6. C_L is not required when performing dc electrical tests F1-F12 and G1-G24.

FIGURE 11. DQK logic load circuit

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TABLE X. Input current (high-level) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H1	Input current, high level	22	Open	24	0	40	µA	12
H2		24		22				
H3		23		2				
H4		2		23				
H5		25		5				
H6		5		25				
H7		6		27				
H8		27		6				
H9		7		8				
H10		8		7				
H11		30		9				
H12		9		30				
H13		32		33				
H14		33		32				
H15		14		31				
H16		31		14				
H17		34		15				
H18		15		34				
H19		36		17				

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TABLE X. Input current (high-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H20	Input current, high level	17	Open	36	0	40	µA	12
H21	↓	19	↓	38	↓	↓	↓	↓
H22		38		19				
H23		18		37				
H24		37		18				

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TABLE XI. Input current (low-level) test.

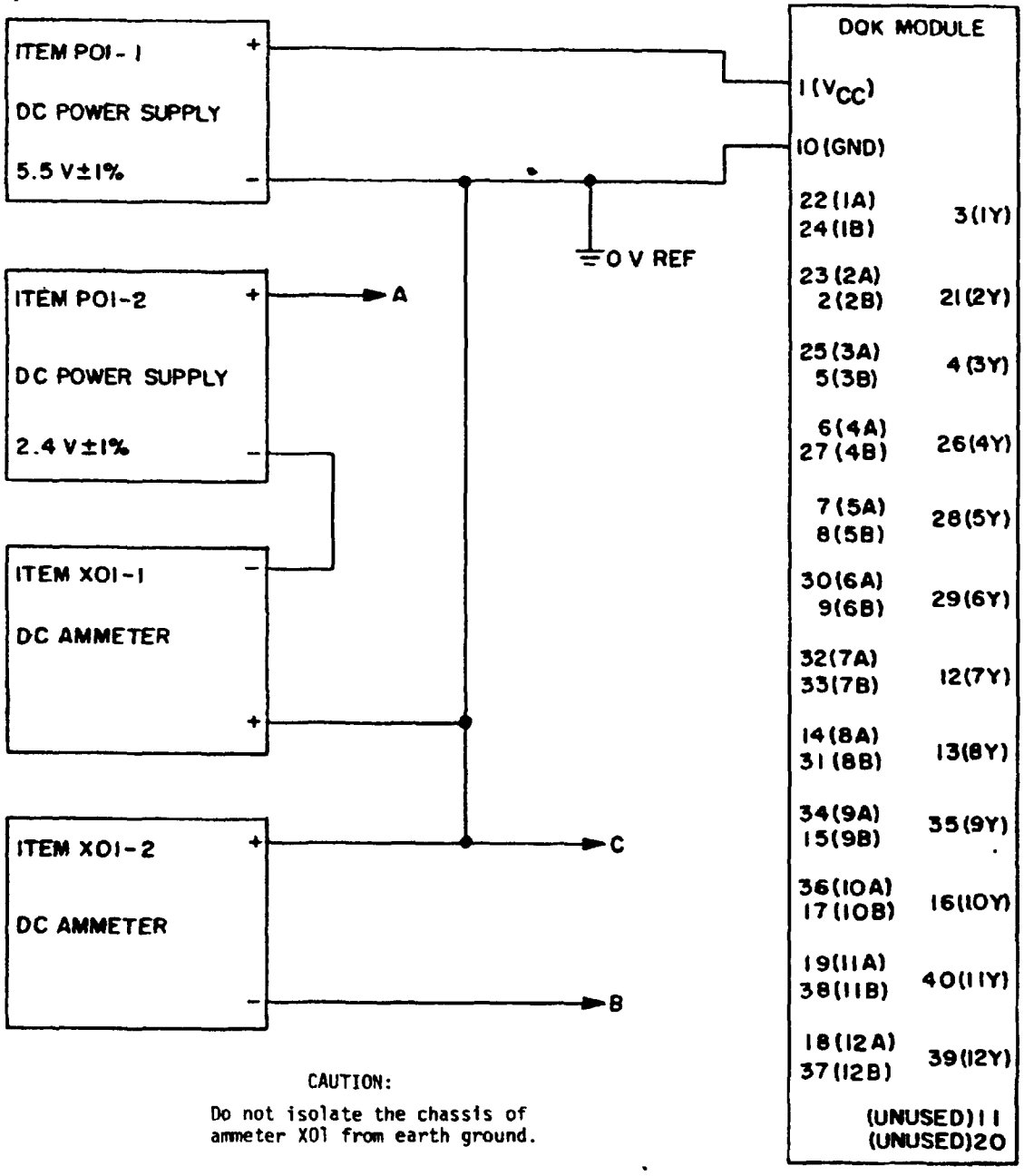
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J1	Input current, low level	Open	22	Open	-0.10	-1.73	mA	12
J2			24					
J3			23					
J4			2					
J5			25					
J6			5					
J7			6					
J8			27					
J9			7					
J10			8					
J11			30					
J12			9					
J13			32					
J14			33					
J15			14					
J16			31					
J17			34					
J18			15					
J19			36					

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TABLE XI. Input current (low-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J20	Input current, low level	Open	17	Open	-0.10	-1.73	mA	12
J21	↓	↓	19	↓	↓	↓	↓	↓
J22			38					
J23			18					
J24			37					

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTE Since items X01-1 and X01-2 are not used simultaneously, only one item X01 is required.

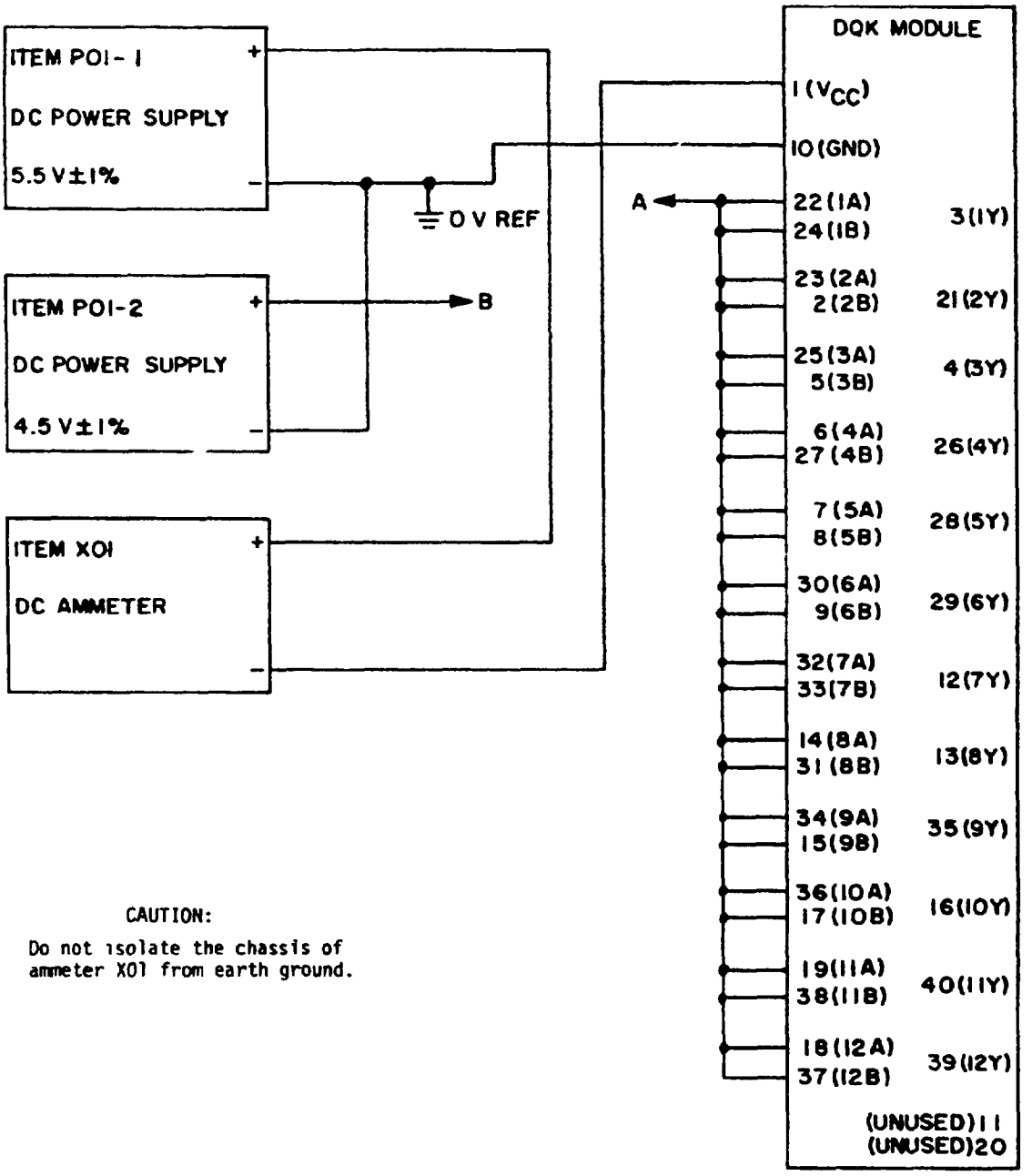
FIGURE 12 Input current (high-level and low-level) test circuit for tables X and XI

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TABLE XII. Power supply current (low-level and high-level) test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
K1	Power supply current, low level	10	1	99	mA	13
K2	Power supply current, high level	B	1	63	mA	13

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CAUTION:
Do not isolate the chassis of ammeter XO1 from earth ground.

FIGURE 13. Power supply current (low-level and high-level) test circuit for table XII.

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TABLE XIII. Transition time (t_{TLH}) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
L1	Transition time (t_{TLH}), C_L = 50 pF	22	3	24,2,5,27,8, 9,33,31,15, 17,38,37	3	45	ns	14, 15
L2		23	21					
L3		25	4					
L4		6	26					
L5		7	28					
L6		30	29					
L7		32	12					
L8		14	13					
L9		34	35					
L10		36	16					
L11		19	40					
L12		18	39					
L13	Transition time (t_{TLH}), C_L = 150 pF	22	3			70		
L14		23	21					
L15		25	4					
L16		6	26					
L17		7	28					

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TABLE XIII. Transition time (t_{TLH}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
L18	Transition time (t_{TLH}), C_L = 150 pF	30	29	24,2,5,27,8, 9,33,31,15, 17,38,37	3	70	ns	14, 15
L19		32	12					
L20		14	13					
L21		34	35					
L22		36	16					
L23		19	40					
L24		18	39					

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TABLE XIV. Transition time (t_{THL}) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
M1	Transition time (t_{THL}), C_L = 50 pF	22	3	24,2,5,27, 8,9,33,31, 15,17,38,37	4	20	ns	14, 15
M2		23	21					
M3		25	4					
M4		6	26					
M5		7	28					
M6		30	29					
M7		32	12					
M8		14	13					
M9		34	35					
M10		36	16					
M11		19	40					
M12		18	39					
M13	Transition time (t_{THL}), C_L = 150 pF	22	3			30		
M14		23	21					
M15		25	4					
M16		6	26					

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TABLE XIV. Transition time (t_{THL}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
M17	Transition time (t_{THL}), C_L = 150 pF	7	28	24,2,5,27, 8,9,33,31, 15,17,38,37	4	30	ns	14, 15
M18	↓	30	29	↓	↓	↓	↓	↓
M19	↓	32	12	↓	↓	↓	↓	↓
M20	↓	14	13	↓	↓	↓	↓	↓
M21	↓	34	35	↓	↓	↓	↓	↓
M22	↓	36	16	↓	↓	↓	↓	↓
M23	↓	19	40	↓	↓	↓	↓	↓
M24	↓	18	39	↓	↓	↓	↓	↓

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TABLE XV. Propagation delay time (t_{PLH}) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
N1	Propagation delay time (t_{PLH}), C_L = 50 pF	22	3	24,2,5,27, 8,9,33,31, 15,17,38,37	3	31	ns	14, 15
N2		23	21					
N3		25	4					
N4		6	26					
N5		7	28					
N6		30	29					
N7		32	12					
N8		14	13					
N9		34	35					
N10		36	16					
N11		19	40					
N12		18	39					
N25	Propagation delay time (t_{PLH}), C_L = 150 pF	22	3			41		

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TABLE XV. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
N26	Propagation delay time (t_{PLH}), C_L = 150 pF	23	21	24,2,5,27, 8,9,33,31, 15,17,38,37	3	41	ns	14, 15
N27		25	4					
N28		6	26					
N29		7	28					
N30		30	29					
N31		32	12					
N32		14	13					
N33		34	35					
N34		36	16					
N35		19	40					
N36		18	39					
N13	Propagation delay time (t_{PLH}), C_L = 50 pF	24	3	22,23,25,6, 7,30,32,14, 34,36,19,18		31		
N14		2	21					
N15		5	4					
N16		27	26					

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TABLE XV. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
N17	Propagation delay time (t_{PLH}), C_L = 50 pF	8	28	22,23,25,6, 7,30,32,14, 34,36,19,18	3	31	ns	14, 15
N18		9	29					
N19		33	12					
N20		31	13					
N21		15	35					
N22		17	16					
N23		38	40					
N24		37	39					
N37	Propagation delay time (t_{PLH}), C_L = 150 pF	24	3			41		
N38		2	21					
N39		5	4					
N40		27	26					
N41		8	28					
N42		9	29					
N43		33	12					
N44		31	13					

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TABLE XV. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
N25	Propagation delay time (t_{PLH}), C_L = 150 pF	15	35	22,23,25,6, 7,30,32,14, 34,36,19,18	3	41	ns	L4, 15
N46	↓	17	16	↓	↓	↓	↓	↓
N47		38	40					
N48		37	39					

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TABLE XVI. Propagation delay time (t_{PHL}) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
P1	Propagation delay time (t_{PHL}), C_L = 50 pF	22	3	24,2,5,27,8 9,33,31,15 17,38,37	3	22	ns	14, 15
P2		23	21					
P3		25	4					
P4		6	26					
P5		7	28					
P6		30	29					
P7		32	12					
P8		14	13					
P9		34	35					
P10		36	16					
P11		19	40					
P12		18	39					
P25	Propagation delay time (t_{PHL}), C_L = 150 pF	22	3			32		
P26		23	21					
P27		25	4					
P28		6	26					

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TABLE XVI. Propagation delay time (t_{pHL}) test - Continued.

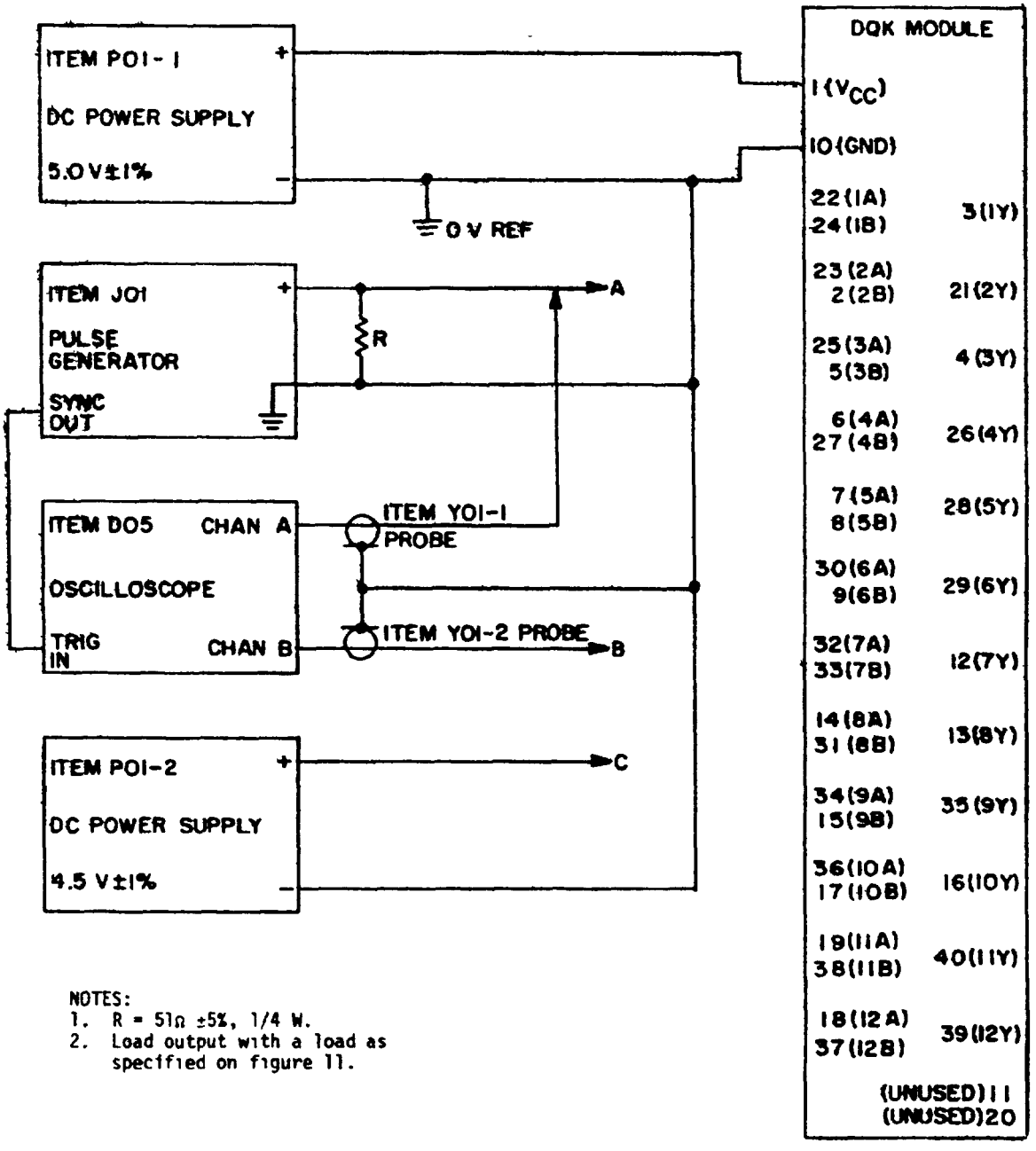
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
P29	Propagation delay time (t_{pHL}), C_L = 150 pF	7	28	24,2,5,27, 8,9,33,31, 15,17,38,37	3	32	ns	14, 15
P30		30	29					
P31		32	12					
P32		14	13					
P33		34	35					
P34		36	16					
P35		19	40					
P36		18	39					
P13	Propagation delay time (t_{pHL}), C_L = 50 pF	24	3	22,23,25,6, 7,30,32,14, 34,36,19,18		22		
P14		2	21					
P15		5	4					
P16		27	26					
P17		8	28					
P18		9	29					
P19		33	12					

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TABLE XVI. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
P20	Propagation delay time (t_{PHL}), C_L = 50 pF	31	13	22,23,25,6, 7,30,32,14 34,36,19,18	3	22	ns	14, 15
P21		15	35					
P22		17	16					
P23		38	40					
P24		37	39					
P37	Propagation delay time (t_{PHL}), C_L = 150 pF	24	3			32		
P38		2	21					
P39		5	4					
P40		27	26					
P41		8	28					
P42		9	29					
P43		33	12					
P44		31	13					
P45		15	35					
P46		17	16					
P47		38	40					
P48		37	39					

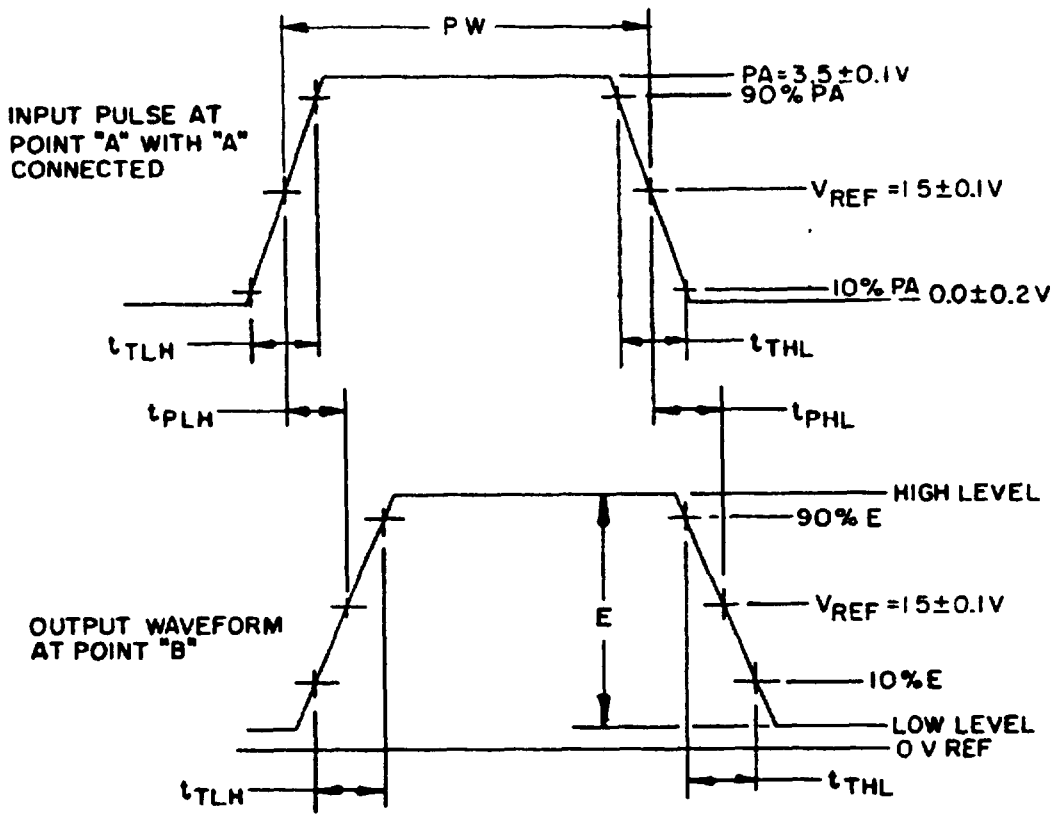
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:54Z
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NOTES:
1. R = 51Ω ±5%, 1/4 W.
2. Load output with a load as specified on figure 11.

FIGURE 14 Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XIII, XIV, XV and XVI.

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NOTES:
1. Input pulse parameters:
   PW = 100 ns ±10%.
   PRR = 5 MHZ ±10%.
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
2. Verify that the output has a high-level $\geq$ 2.4 V and low-level $\leq$ 0.4 V before making measurements.

FIGURE 15. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XIII, XIV, XV and XVI.

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TABLE XVII. Output crosstalk (high-level) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q1	Output crosstalk, high level	22	Open	23,25,6,7,30, 32,14,34,36, 19,18	3	0	400	mV	16,17
Q2		23		22,25,6,7,30, 32,14,34,36, 19,18	21				
Q3		25		22,23,6,7,30, 32,14,34,36, 19,18	4				
Q4		6		22,23,25,7,30 32,14,34,36, 19,18	26				
Q5		7		22,23,25,6,30 32,14,34,36, 19,18	28				
Q6		30		22,23,25,6,7, 32,14,34,36, 19,18	29				
Q7		32		22,23,25,6,7, 30,14,34,36, 19,18	12				
Q8		14		22,23,25,6,7, 30,32,34,36, 19,18	13				

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Check the source to verify that this is the current version before use.

TABLE XVII. Output crosstalk (high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q9	Output crosstalk, high level	34	Open	22,23,25,6,7,30,32,14,36,19,18	35	0	400	mV	16,17
Q10		36		22,23,25,6,7,30,32,14,34,19,18	16				
Q11		19		22,23,25,6,7,30,32,14,34,36,18	40				
Q12		18		22,23,25,6,7,30,32,14,34,36,19	39				

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Check the source to verify that this is the current version before use.

* TABLE XVIII. Output crosstalk (low-level) test.

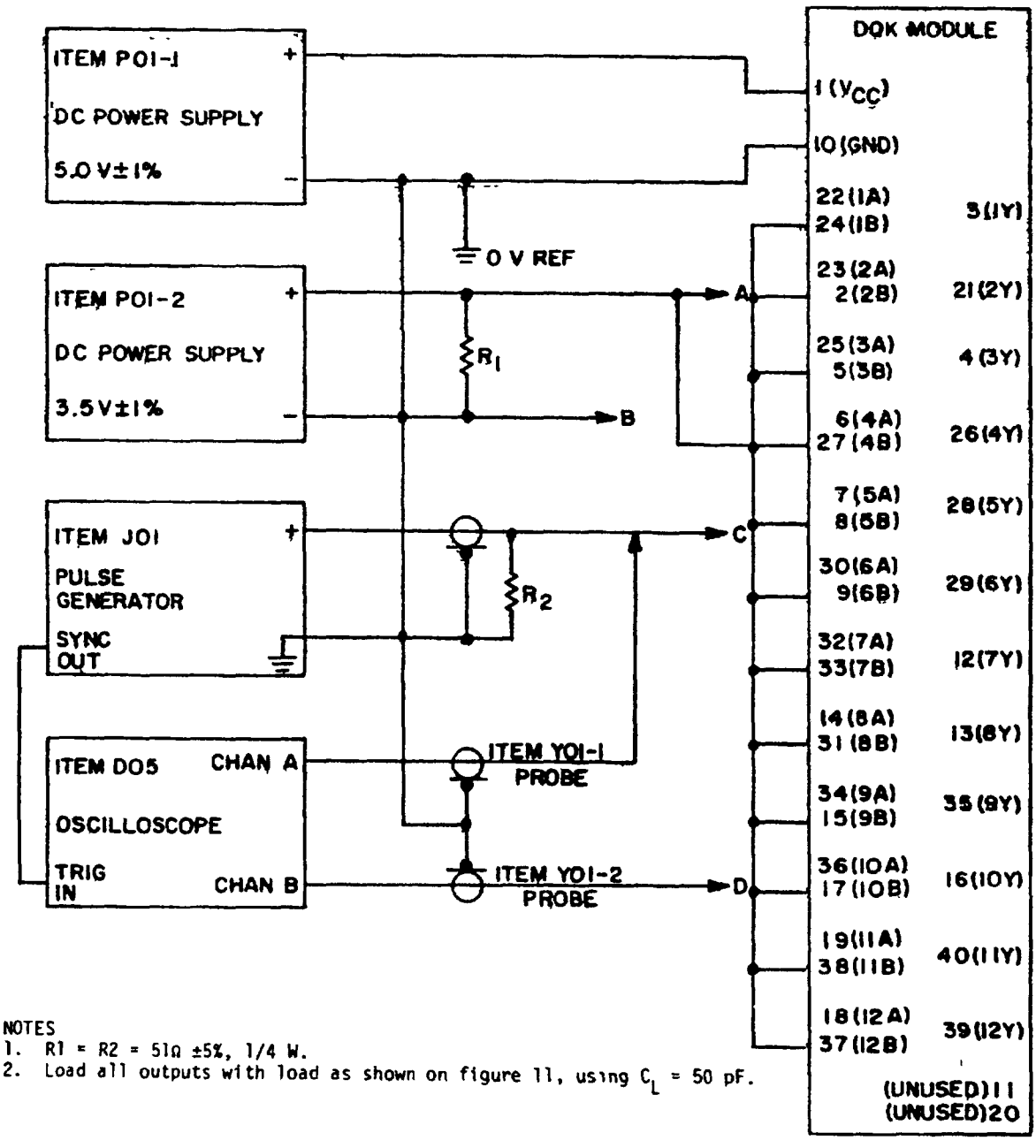
Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	Output crosstalk, low level	Open	22	23,25,6,7,30,32,14,34,36,19,18	3	0	800	mV	16,17
R2			23	22,25,6,7,30,32,14,34,36,19,18	21				
R3			25	22,23,6,7,30,32,14,34,36,19,18	4				
R4			6	22,23,25,7,30,32,14,34,36,19,18	26				
R5			7	22,23,25,6,30,32,14,34,36,19,18	28				
R6			30	22,23,25,6,7,32,14,34,36,19,18	29				
R7			32	22,23,25,6,7,30,14,34,36,19,18	12				
R8			14	22,23,26,6,7,30,32,34,36,19,18	13				
R9			34	22,23,25,6,7,30,32,14,36,19,18	35				

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* TABLE XVIII. Output crosstalk (low-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R10	Output crosstalk, low level	Open	36	22,23,25,6,7,30,32,14,34,19,18	16	0	800	mV	16,17
R11	↓	↓	19	22,23,25,6,7,30,32,14,34,36,18	40	↓	↓	↓	↓
R12	↓	↓	18	22,23,25,6,7,30,32,14,34,36,19	39	↓	↓	↓	↓

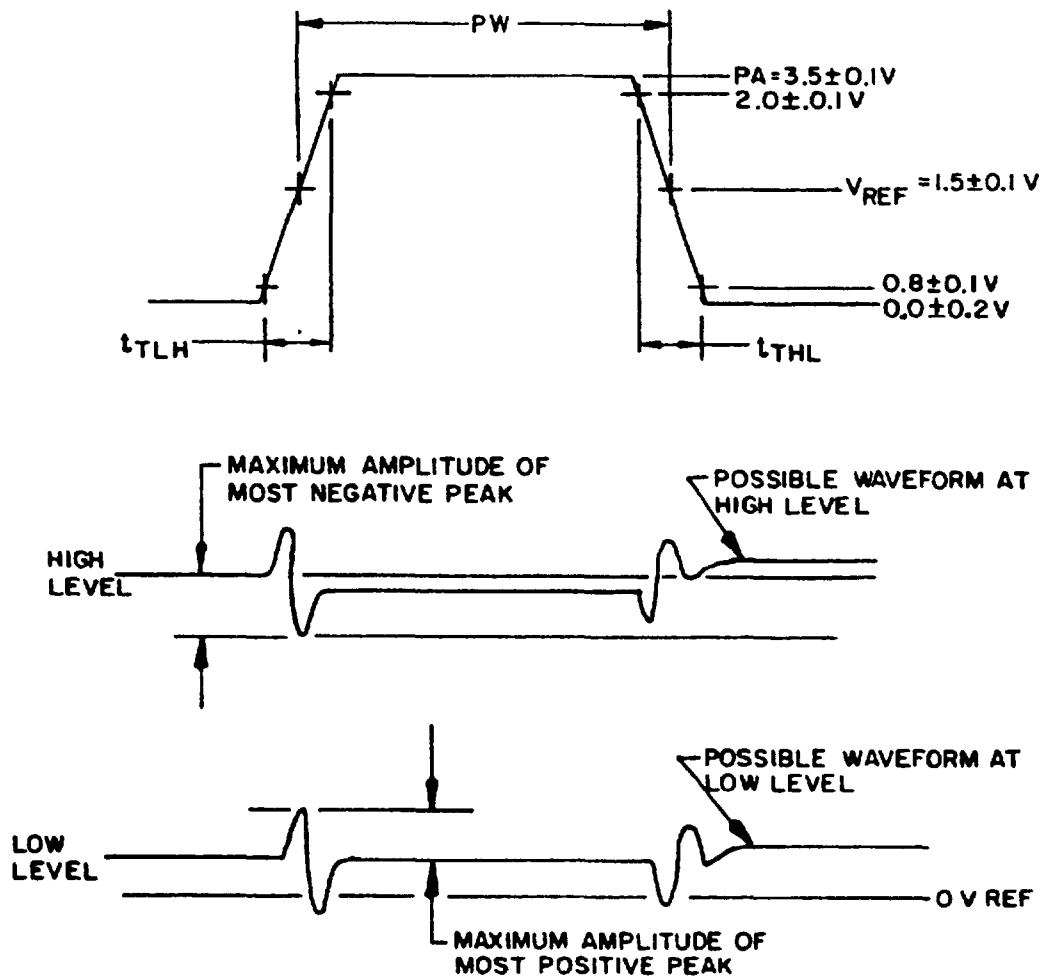
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NOTES
1. R1 = R2 = 51Ω ±5%, 1/4 W.
* 2. Load all outputs with load as shown on figure 11, using C_L = 50 pF.

FIGURE 16 Output crosstalk (high-level and low-level) test circuit for tables XVII and XVIII.

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NOTE: Input pulse parameters:
PW = 200 ns ±10%.
$t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns verified at each input pin.
PRR = 1 MHz ±10%.

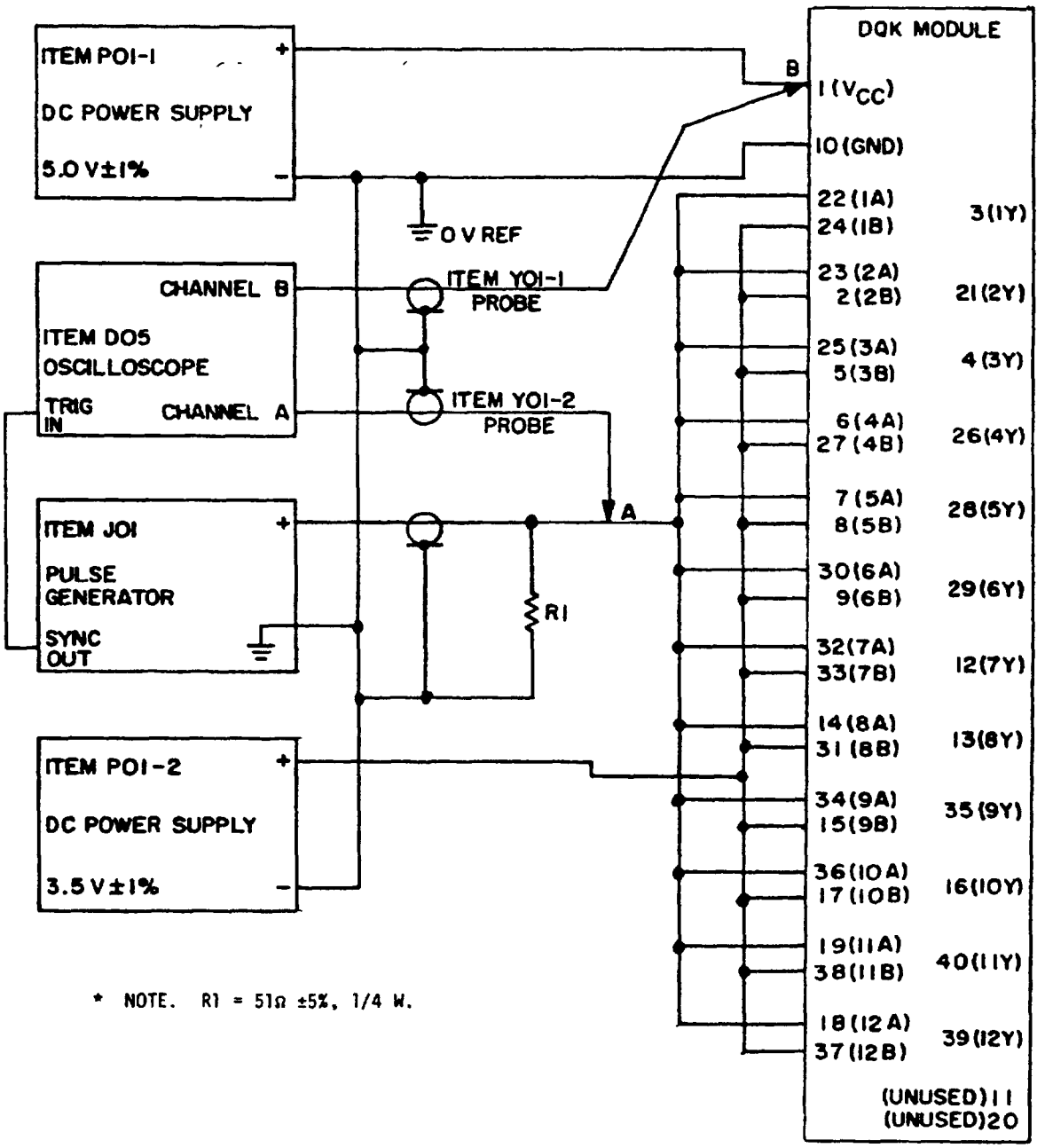
FIGURE 17. Output crosstalk (high-level and low-level) test circuit waveforms for tables XVII and XVIII.

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TABLE XIX. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
S	Power supply transient amplitude	0	400	mV	18, 19

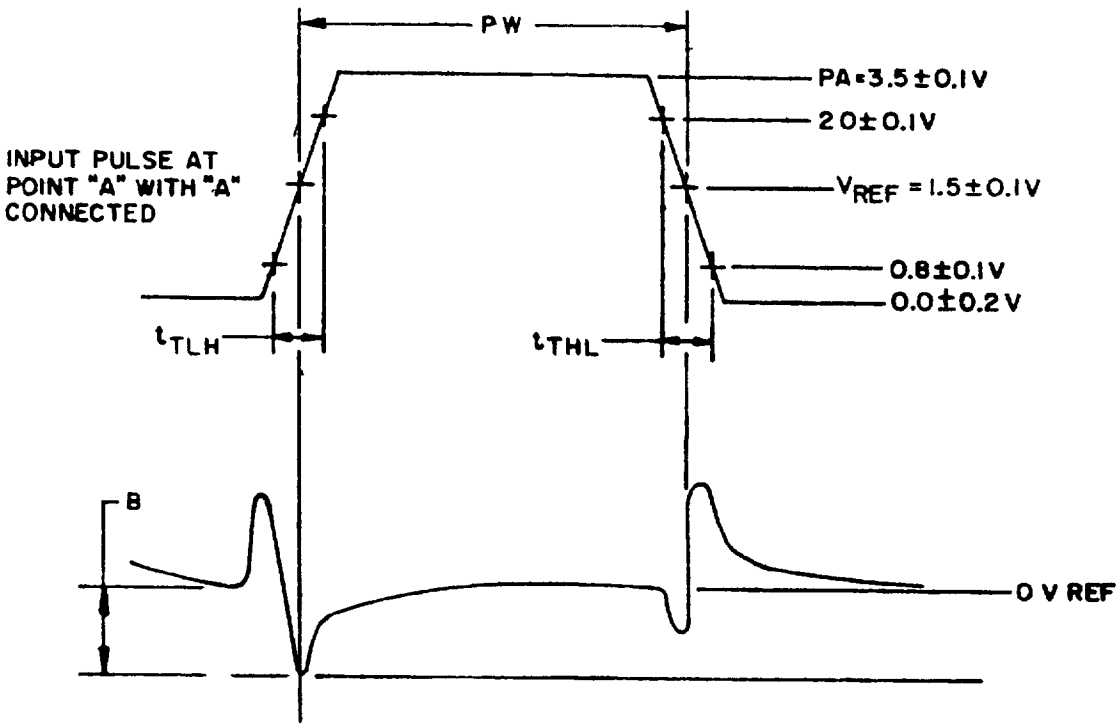
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* NOTE. R1 = 51Ω ±5%, 1/4 W.

FIGURE 18 Power supply transient amplitude test circuit for table XIX

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* NOTES
1. Input pulse parameters:
   PW = 3 µs ±10%
   $t_{TLH} = t_{THL}$ = 3.5 ±1 5 ns verified at each input pin.
   PRR = 100 kHz ±10%.
2. B = Measurement from the baseline to most negative excursion of the waveform.

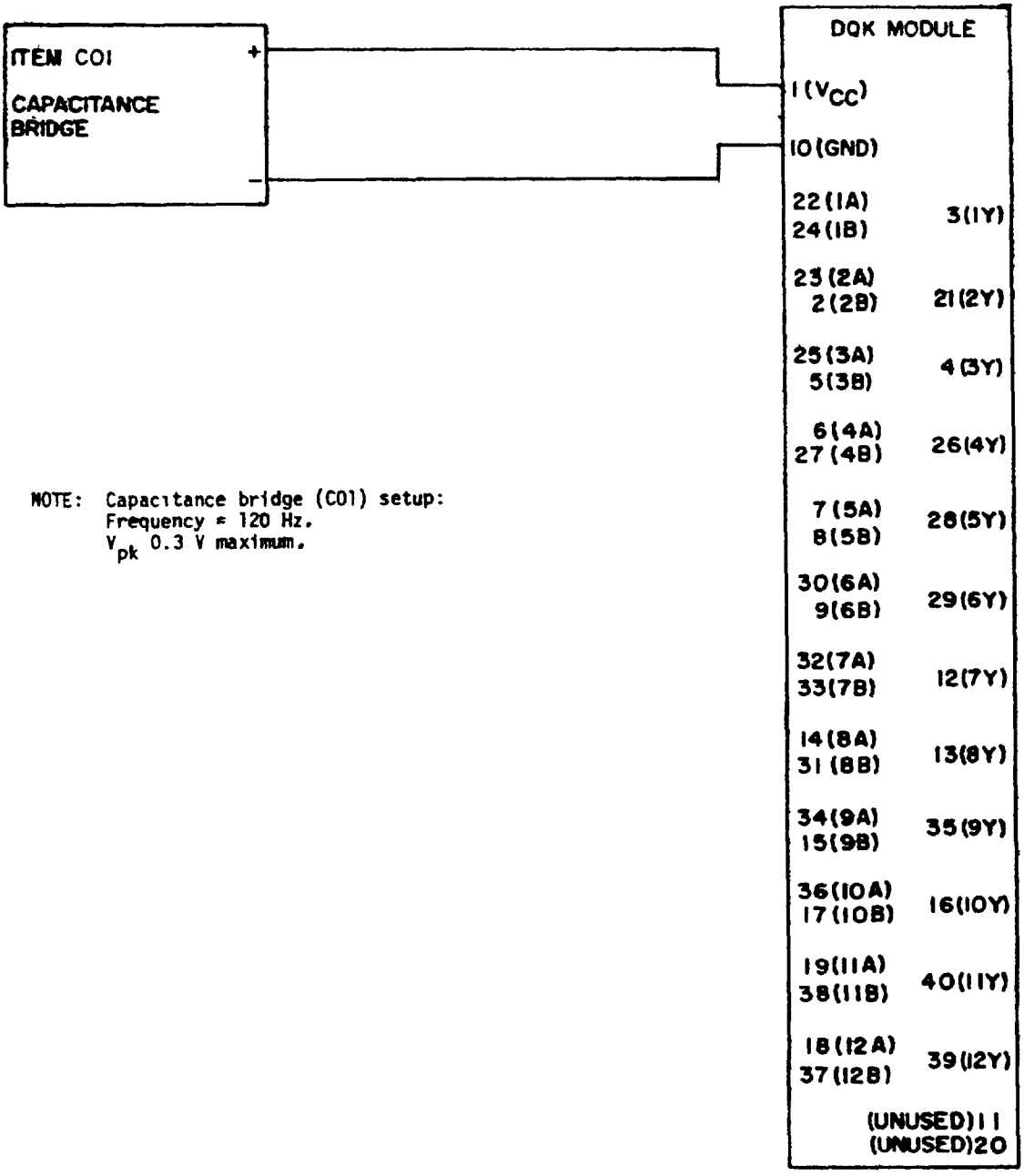
FIGURE 19 Power supply transient amplitude test circuit waveforms for table XIX.

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TABLE XX. Power supply filter capacitance test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
T	Power supply filter capacitance	1.6	3.0	µF	20

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NOTE: Capacitance bridge (CO1) setup:
Frequency = 120 Hz.
V_{pk} 0.3 V maximum.

FIGURE 20. Power supply filter capacitance test circuit for table XX.

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TABLE XXI. Input capacitance test.

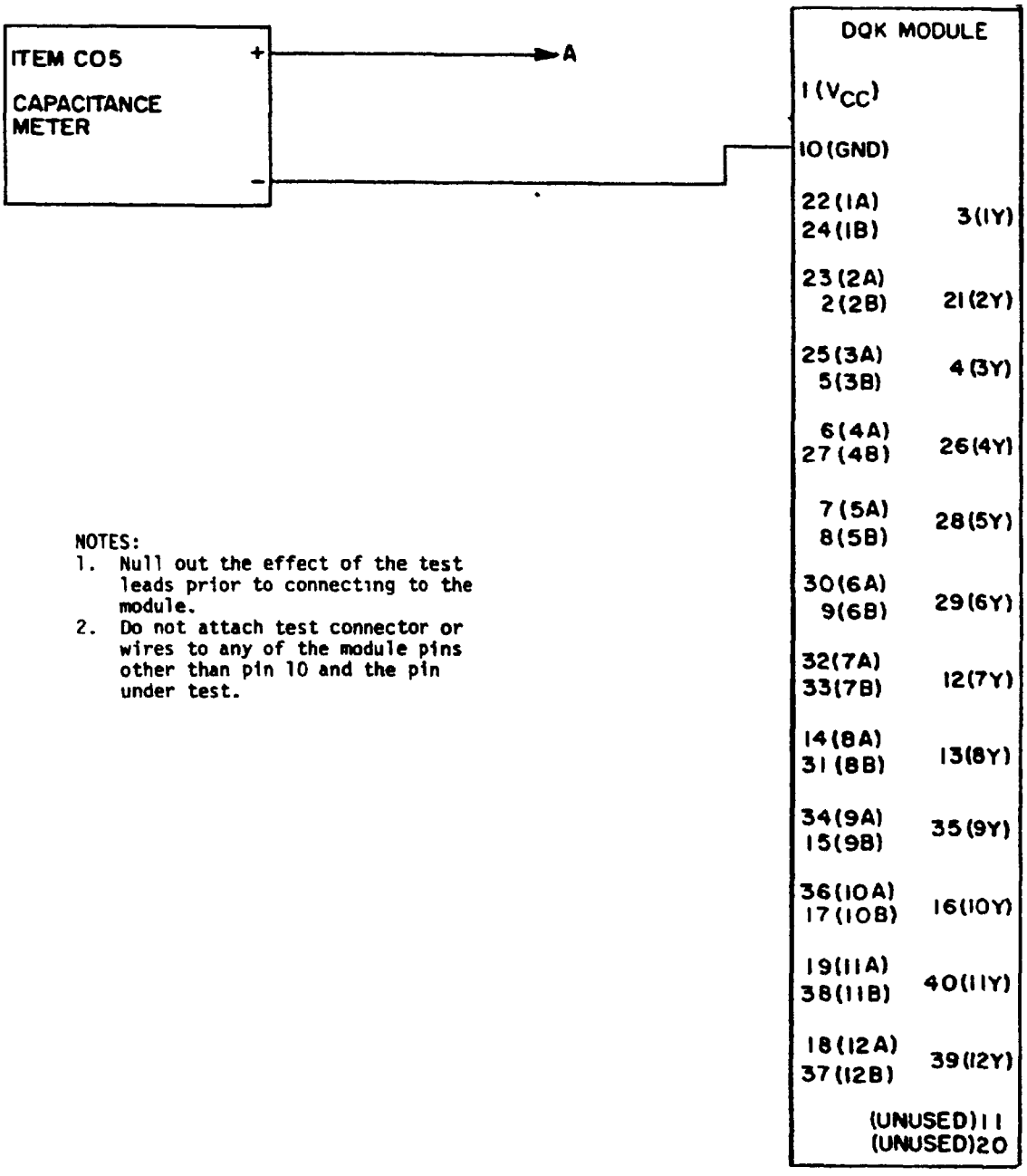
Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
U1	Input capacitance	22	2	15	pF	21
U2		24				
U3		23				
U4		2				
U5		25				
U6		5				
U7		6				
U8		27				
U9		7				
U10		8				
U11		30				
U12		9				
U13		32				
U14		33				
U15		14				
U16		31				
U17		34				
U18		15				
U19		36				

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TABLE XXI. Input capacitance test - Continued.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
U20	Input capacitance	17	2	15	pF	21
U21	↓	19	↓	↓	↓	↓
U22		38				
U23		18				
U24		37				

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NOTES:
1. Null out the effect of the test leads prior to connecting to the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test.

FIGURE 21 Input capacitance test circuit for table XXI.

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